KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5281 | 5 Credits: 4 | Program Core | Source-grounded

Object Oriented Programming

- Understand object-oriented Principles.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5281 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5281.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Computer Engineering / Computer Hardware Engineering/
Course code	5281
Course title	Object Oriented Programming
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

31 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● Understand object-oriented Principles.
- ● Write, Compile and Execute Java Programs.
- ● Familiarize GUI programming using Swing.
- ● Introduce database connectivity for developing applications.

Course outcomes

CO	Official outcome	Study level
CO1	16 Applying develop simple JAVA applications. Build classes using inheritance, interfaces and	See official syllabus
CO2	16 Applying packages	See official syllabus
CO3	Build GUI applications using swing. 14 Applying	See official syllabus
CO4	Develop applications using Database. 12 Applying	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	applications	12	As specified
Module 2	Build classes using inheritance, interfaces and packages.	8	As specified
Module 3	Build GUI applications using swing.	6	As specified
Module 4	Develop applications using database	5	As specified

MODULE 1

applications

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: applications
- Official duration: As specified
- Official topic entries captured: 12

- The full source excerpt is preserved below for coverage checking.

1 Understanding. Programming. Explain the JDK & IDE’s for developing java

1 Understanding. Programming. Explain the JDK & IDE’s for developing java is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. Programming. Explain the JDK & IDE’s for developing java using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. programming. explain the jdk & ide’s for developing java should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. Programming. Explain the JDK & IDE’s for developing java means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 1 Understanding. Programming. Explain the JDK & IDE’s for developing java [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding. applications Outline the general structure of a Java Understanding.

1 Understanding. applications Outline the general structure of a Java Understanding. is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. applications Outline the general structure of a Java Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. applications outline the general structure of a java understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

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Definition/identity	What 1 Understanding. applications Outline the general structure of a Java Understanding. means in the official course context
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program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes

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Study points

- Define or identify program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, program and explain steps to create, compile 1 and execute a program describe objects and classes and use classes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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2 Understanding. to model objects. Demonstrate how to define classes and create

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. to model objects. Demonstrate how to define classes and create using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. to model objects. demonstrate how to define classes and create should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. to model objects. Demonstrate how to define classes and create means in the official course context
Study lens	Principle → components or stages → result → application
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1 Applying. objects

1 Applying. objects is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 1 Applying. objects using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying. objects should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying. objects means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Create Objects using constructors. 1 Applying Interpret accessing objects via object

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Create Objects using constructors. 1 Applying Interpret accessing objects via object using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, create objects using constructors. 1 applying interpret accessing objects via object should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Create Objects using constructors. 1 Applying Interpret accessing objects via object means in the official course context
Study lens	Principle → components or stages → result → application
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1 Understanding. reference variables. Distinguish between instance and static

1 Understanding. reference variables. Distinguish between instance and static is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify 1 Understanding. reference variables. Distinguish between instance and static using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, 1 understanding. reference variables. distinguish between instance and static should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. reference variables. Distinguish between instance and static means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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1 Understanding. variables and methods. Develop Java Programs to implement classes

1 Understanding. variables and methods. Develop Java Programs to implement classes is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 1 Understanding. variables and methods. Develop Java Programs to implement classes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
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Engineering or professional relevance

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2 Applying. and objects. Explain Visibility Modifiers and Data

2 Applying. and objects. Explain Visibility Modifiers and Data is an official topic in Module 1 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 2 Applying. and objects. Explain Visibility Modifiers and Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying. and objects. explain visibility modifiers and data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

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Definition/identity	What 2 Applying. and objects. Explain Visibility Modifiers and Data means in the official course context
Study lens	Principle → components or stages → result → application
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1 Understanding. Encapsulation Develop methods with object arguments an

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Study points

- Define or identify 1 Understanding. Encapsulation Develop methods with object arguments an using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. encapsulation develop methods with object arguments an should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. Encapsulation Develop methods with object arguments an means in the official course context
Study lens	Principle → components or stages → result → application
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differentiate between Primitive arguments 2 Applying and object type arguments

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Study points

- Define or identify differentiate between Primitive arguments 2 Applying and object type arguments using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, differentiate between primitive arguments 2 applying and object type arguments should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What differentiate between Primitive arguments 2 Applying and object type arguments means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-[illegible] differentiate between Primitive arguments 2 Applying and object type arguments [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement.

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Study points

- Define or identify Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain exception handling. 2 understanding contnets: object oriented programming (oop) – characteristics of oop – features of java – advantages of java – tools available of java programming (jdk, java packages, various ides like netbeans, eclipse) – building java applications. objects and classes – defining a class – declaring attributes, declaring and defining methods, creating object – accessing objects - constructors – constructor overloading – static variables, constants and methods – method overloading – visibility modifiers, data field encapsulation, passing and returning objects as arguments, array of objects-exception handling, try,catch,-multiple catch & finally statement. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain Exception Handling. 2 Understanding Contnets: Object Oriented Programming (OOP) – Characteristics of OOP – Features of JAVA – Advantages of JAVA – Tools Available of JAVA Programming (JDK, JAVA Packages, various IDEs like NetBeans, eclipse) – Building Java applications. Objects and Classes – Defining a Class – Declaring attributes, Declaring and Defining methods, Creating Object – Accessing Objects - Constructors – Constructor overloading – Static variables, constants and methods – method overloading – Visibility modifiers, Data field Encapsulation, passing and returning objects as arguments, Array of objects-Exception handling, Try,catch,-Multiple catch & finally statement. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
applications
            Explain the features of Object-Oriented
  M1.01                                                        1
Understanding.
            Programming.
            Explain the JDK & IDE’s for developing java
  M1.02                                                        1
Understanding.
            applications
            Outline the general structure of a Java

Understanding.
  M1.03     program and explain steps to create, compile       1
            and execute a program
            Describe objects and classes and use classes
  M1.04                                                        2
Understanding.
            to model objects.
            Demonstrate how to define classes and create
  M1.05                                                        1       Applying.
            objects
  M1.06     Create Objects using constructors.                 1       Applying
            Interpret accessing objects via object
  M1.07                                                        1
Understanding.
            reference variables.
            Distinguish between instance and static
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Build classes using inheritance, interfaces and packages.

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Build classes using inheritance, interfaces and packages.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

3 Understanding inheritance with examples. Illustrate invoking the superclass constructor

3 Understanding inheritance with examples. Illustrate invoking the superclass constructor is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding inheritance with examples. illustrate invoking the superclass constructor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Applying and methods using super keywords. Illustrate overriding instance methods in the

2 Applying and methods using super keywords. Illustrate overriding instance methods in the is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying and methods using super keywords. Illustrate overriding instance methods in the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying and methods using super keywords. illustrate overriding instance methods in the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying and methods using super keywords. Illustrate overriding instance methods in the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Applying and methods using super keywords. Illustrate overriding instance methods in the [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Applying subclass.

2 Applying subclass. is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying subclass. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying subclass. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying subclass. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Applying subclass. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate polymorphism and dynamic binding

Illustrate polymorphism and dynamic binding is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate polymorphism and dynamic binding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate polymorphism and dynamic binding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate polymorphism and dynamic binding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Illustrate polymorphism and dynamic binding [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain visibility controls 1 Understanding Build abstract class, final class and method

Explain visibility controls 1 Understanding Build abstract class, final class and method is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain visibility controls 1 Understanding Build abstract class, final class and method using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain visibility controls 1 understanding build abstract class, final class and method should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain visibility controls 1 Understanding Build abstract class, final class and method means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain visibility controls 1 Understanding Build abstract class, final class and method □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Applying overriding

2 Applying overriding is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying overriding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying overriding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying overriding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying overriding □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Develop interfaces

Develop interfaces is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop interfaces using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop interfaces should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop interfaces means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Develop interfaces □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding - Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages.

Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding - Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. is an official topic in Module 2 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding - Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop packages 2 applying inheritance – types of inheritance – superclass and subclass – calling super class constructor and methods – overloading vs overriding - final variables, final methods and final classes – abstract methods and class – polymorphism – dynamic binding – visibility controls. interfaces – definition –

extending, implementing and accessing interfaces- packages – creating and accessing of packages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding - Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Develop packages 2 Applying Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – overloading vs overriding - Final variables, final methods and final classes – Abstract methods and class – polymorphism – dynamic binding – Visibility controls. Interfaces – Definition – Extending, Implementing and accessing Interfaces- packages – Creating and Accessing of Packages. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Build classes using inheritance, interfaces and packages.
            Describe inheritance and different types of
 M2.01                                                        3
Understanding
            inheritance with examples.
            Illustrate invoking the superclass constructor
 M2.02                                                        2      Applying
            and methods using super keywords.
            Illustrate overriding instance methods in the
 M2.03                                                        2      Applying
            subclass.
 M2.04      Illustrate polymorphism and dynamic binding       2      Applying
 M2.05      Explain visibility controls                       1
Understanding
            Build abstract class, final class and method
 M2.06                                                        2      Applying
            overriding
 M2.07      Develop interfaces                                2      Applying
 M2.08      Develop packages                                  2      Applying
            Series Test I                                     1
Contents:
Inheritance – Types of Inheritance – Superclass and subclass – calling super
class
constructor and methods – overloading vs overriding - Final variables, final
methods and
final classes – Abstract methods and class – polymorphism – dynamic binding –
Visibility
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Build GUI applications using swing.

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Build GUI applications using swing.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI

Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline gui programming in java using swing. 2 understanding. make use of packages containing gui should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications.

containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, containers, components, event-handling classes 2 applying. and interfaces to develop simple applications. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and

Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe steps to create and manipulate gui 3 understanding. controls. explain handling of mouse events and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding. keyboard events. Explain Common GUI event type and Listener

2 Understanding. keyboard events. Explain Common GUI event type and Listener is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. keyboard events. Explain Common GUI event type and Listener using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. keyboard events. explain common gui event type and listener should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. keyboard events. Explain Common GUI event type and Listener means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding. keyboard events. Explain Common GUI event type and Listener □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Understanding. types.

2 Understanding. types. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. types. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 2 Understanding. types. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as Jlabel, JtextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea – Event Handling – Common GUI event Types and Containers.

Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as Jlabel, JtextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea – Event Handling – Common GUI event Types and Containers. is an official topic in Module 3 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as Jlabel, JtextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea – Event Handling – Common GUI event Types and Containers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs using swing components 3 applying. outline gui programming – swing – swing components – simple gui based input/output programs - gui application development using swing components such as jlabel, jtextfield, jbutton, jlist, jcheckbox, jcombobox, jpanel, jtextarea – event handling – common gui event types and containers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as Jlabel, JtextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea – Event Handling – Common GUI event Types and Containers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Develop programs using swing Components 3 Applying. Outline GUI programming – Swing – Swing Components – Simple GUI based input/output programs - GUI application development using Swing components such as Jlabel, JtextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea – Event Handling – Common GUI event Types and Containers. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Build GUI applications using swing.
 M3.01      Outline GUI Programming in Java using swing.          2
Understanding.
            Make use of packages containing GUI
 M3.02      containers, components, event-handling classes        2
Applying.
            and interfaces to develop simple applications.
 M3.03      Describe steps to create and manipulate GUI           3
Understanding.
            controls.

            Explain handling of mouse events and
 M3.04                                                            2
Understanding.
            keyboard events.
            Explain Common GUI event type and Listener
 M3.05                                                            2
Understanding.
            types.
 M3.06      Develop programs using swing Components               3
Applying.

Contents:
Outline GUI programming – Swing – Swing Components – Simple GUI based
input/output programs - GUI application development using Swing components such
as
Jlabel, JtextField, Jbutton, Jlist, Jcheckbox, Jcombobox, Jpanel, Jtextarea –
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Develop applications using database

This module is presented from the official syllabus scope for Object Oriented Programming. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop applications using database
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Explain relational database & SQL

Explain relational database & SQL is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain relational database & SQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain relational database & sql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain relational database & SQL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain relational database & SQL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain SQL commands.

Explain SQL commands. is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain SQL commands. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sql commands. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain SQL commands. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain SQL commands. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the

Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the stages in jdbc 2 understanding illustrate the interfaces in jdbc –register the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-□□ Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset

driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, driver- connection- preparedstatement- 2 applying executing statements- resultset should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java

Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java is an official topic in Module 4 of Object Oriented Programming. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop simple application with database 4 applying database – creation of databases-tables -sql commands like -create, alter, drop, select, insert, delete database connectivity – jdbc – rowset interface – preparedstatement - retrieving results - closing connection. text / reference t/r book title/author herbert schildt, dale skrien: java fundamentals a comprehensive t1 introduction t2 balaguruswamy e: programming with java, 6th edition. r1 liang, y daniel: introduction to java programming, pearson, 9th ed. r2 herbert schildt: java the complete reference, seventh edition, r3 herbert schildt: swing a beginner's guide r4 paul deital, harvey deital: java how to program, r5 yang hu: easy learning jdbc+mysql online resources sl.no website link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner's guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-□□ Develop simple application with database 4 Applying Database – Creation of databases-Tables -SQL commands like -create, alter, drop, select, insert, delete Database Connectivity – JDBC – RowSet interface – PreparedStatement - retrieving results - closing connection. Text / Reference T/R Book Title/Author Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive T1 Introduction T2 Balaguruswamy E: Programming with Java, 6th edition. R1 Liang, Y Daniel: Introduction to JAVA Programming, Pearson, 9th Ed. R2 Herbert Schildt: Java The Complete Reference, Seventh Edition, R3 Herbert Schildt: Swing A Beginner’s guide R4 Paul Deital, Harvey Deital: Java How to Program, R5 Yang HU: Easy Learning JDBC+Mysql Online Resources Sl.No Website Link 1 https://onlinecourses.nptel.ac.in/noc20_cs08/course 2 https://www.tutorialspoint.com/java □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Develop applications using database
 M4.01      Explain relational database & SQL                     2
Understanding
 M4.02      Explain SQL commands.                                 2
Understanding
 M4.03      Illustrate the stages in JDBC                         2
Understanding
            Illustrate the interfaces in JDBC –register the
 M4.04      driver- connection- Preparedstatement-                2
Applying
            Executing statements- Resultset
 M4.05      Develop simple application with database              4      Applying
            Series Test – II                                      1

Contents:
Database – Creation of databases-Tables -SQL commands like -create, alter, drop,
select,
insert, delete Database Connectivity – JDBC – RowSet interface –
PreparedStatement -
retrieving results - closing connection.

Text / Reference

 T/R                                      Book Title/Author
         Herbert Schildt, Dale Skrien: Java Fundamentals A Comprehensive
  T1
         Introduction
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding. Programming. Explain the JDK & IDE's for developing java. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. Programming. Explain the JDK & IDE's for developing java*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding. applications Outline the general structure of a Java Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. applications Outline the general structure of a Java Understanding.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes. (3 marks)

Answer: Begin with the standard definition or identity of *program and explain steps to create, compile 1 and execute a program Describe objects and classes and use classes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding. to model objects. Demonstrate how to define classes and create. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. to model objects. Demonstrate how to define classes and create*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 3 Understanding inheritance with examples. Illustrate invoking the superclass constructor. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding inheritance with examples. Illustrate invoking the superclass constructor*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Applying and methods using super keywords. Illustrate overriding instance methods in the. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying and methods using super keywords. Illustrate overriding instance methods in the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Applying subclass.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying subclass.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate polymorphism and dynamic binding. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate polymorphism and dynamic binding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI. (3 marks)

Answer: Begin with the standard definition or identity of *Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications.. (3 marks)

Answer: Begin with the standard definition or identity of *containers, components, event-handling classes 2 Applying. and interfaces to develop simple applications.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and. (3 marks)

Answer: Begin with the standard definition or identity of *Describe steps to create and manipulate GUI 3 Understanding. controls. Explain handling of mouse events and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding. keyboard events. Explain Common GUI event type and Listener. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding. keyboard events. Explain Common GUI event type and Listener*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain relational database & SQL. (3 marks)

Answer: Begin with the standard definition or identity of *Explain relational database & SQL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Explain SQL commands.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain SQL commands.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the stages in JDBC 2 Understanding Illustrate the interfaces in JDBC –register the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset. (3 marks)

Answer: Begin with the standard definition or identity of *driver- connection- Preparedstatement- 2 Applying Executing statements- Resultset*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding. Programming. Explain the JDK & IDE’s for developing java**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding inheritance with examples. Illustrate invoking the superclass constructor**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Outline GUI Programming in Java using swing. 2 Understanding. Make use of packages containing GUI**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain relational database & SQL**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.

Term	Meaning in this lesson
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://onlinecourses.nptel.ac.in/noc20_cs08/course https://www.tutorialspoint.com/java

IN-PAGE SEARCH

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